

PAUL NAMALOMBA

🎓 **MSc(Eng) Structural, BSc(Hons.) Civil**

📍 36 Melville Road, Ottery East, Cape Town, South Africa, 7808

☎ +27637478981 | ✉ paulkbnamalomba@gmail.com

🌐 <https://paulnamalomba.github.io> | [in](#) Paul Namalomba



RESUME SUMMARY

- Over **4 years of total technical experience** in Computational science and Digital infrastructure initiatives in Civil Engineering research and teaching at UCT, including end-to-end deployment of Computational Modelling software, usage of HPC clusters (Slurm, Rocky Linux). Further industry experience in building APIs in C#, Python and Javascript, Data Engineering and Database Management with PostgreSQL + Django ORM, including innovating in data updates with MPI + hyperthreading. My experience breaks down as:
- **2.5 years and counting** of project maintenance, and full-stack development of **SESKA finite-element computational suite** written in GNU C++, Fortran and GiD. First of its kind port from Linux to Windows whilst maintaining powerful Multiprocessing (Hyperthreading) compute methods.
- **5 months experience** developing industry-grade APIs in **C#** and **Javascript**, developing **Flutter** apps for Android, including a new ride-sharing app called Ecoride.
- **Nearly 2 (15 months) years' experience** architecting, maintaining, and optimizing the **SEAT EBE enrollment platform** using Python Django with HTML, PostgreSQL, and LDAP for Security Management – We protected against SQL injections by using the Python Django layer for additional security. We innovated with data loads for up to 30 million rows, by using MPI + hyperthreading, and batch-chunked memory management for effective server resource management at update time. Concept: [Batch Threading And Mpi For Large Db Ops - Paul Namalomba](#)
- Over **750 hours of active training** in Data Science, Data Engineering, High Performance Computing and Back-End Development.
- Led cross-functional teams of up to five teaching assistants in **lecturing** undergraduate students—aligning curriculum with emerging technologies and strategic learning objectives.
- Built my own python data processing packages utilizing Computer Vision for Visual Data Analysis with Google's Gemma AI models, Random Forest outlier detection and other machine-learning innovations. (
- My entire portfolio is available online at m website: (<https://paulnamalomba.github.io>). Where I also publish guides and sample scripts.

POINTS OF REFERENCE

- My entire portfolio is available online at my website: (<https://paulnamalomba.github.io>)
- At my website I maintain guides on technical implementation and system file server management, as well as a history of all the technologies I have worked on **Web-Apps, Android Apps, Static Websites, Dashboards** and **APIs**.
- My proof of qualification is attached to this application.

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CORE TECHNICAL SKILLS

- **Programming Languages:** C++, C#, React Typescript, Python, SQL, Flutter, HTML, Javascript (Nuxt, Node)
- **Web Scraping, Data Science and Statistical Data Analysis:** Python, Jupyter{*}, Microsoft Power BI, Microsoft Tableau, Microsoft Excel, Python RESTful APIs
- **RDBMS, ETL Pipelines and Big Data (Cloud Computing):** Azure Database with PostgreSQL, MySQL, SQLite, MongoDB, REDIS and Cloudflare R2 Object Storage.
- **Machine-Level Computer Programming:** Microsoft VSC++, GNU Autotools, Cygwin64, Mingw32/64 Cross-Compat Tools
- **Programming IoT/Realtime Devices and REST APIs:** Arduino-based MCUs, API programming with C# .NET and Python Django, SignalR Websockets
- **DevOps and CI/CD:** GitHub, Bitbucket, Vercel, DockerHub and Cloudflare.
- **High-Performance Computing and Parallel Computing:** HPC, Slurm Workload Manager, OpenMPI, PETSc, METIS.
- **Data Security Policies:** LDAP Configuration, BCrypt, JWT Bearer
- **Computational Modelling and Analysis:** SESKA, GiD, ABAQUS
- **Machine Learning and Regression Analysis:** Scikit-Learn, TensorFlow, statsmodels, datashadric
- **Advanced Shell Scripting, File System and Server Management and Automation:** Windows CMD Prompt, PowerShell 7+, Git BASH, MSYS2 Shells, Linux BASH.

OTHER TECHNICAL+RESEARCH SKILLS

- **Scientific Research Design, Reporting and Methodology**
- **The Engineering Design Process Workflow**
- A Multi-Disciplinary Depth of Knowledge (**Computing, Data Science and Statistics, Engineering and Research Management**)
- **Graphics Design for Posters:** Adobe Photoshop, Canva.
- **Other Advanced ICT Skills:** Microsoft Office, LibreOffice, Microsoft OneNote.
- **Operating Systems:** Linux Ubuntu 20.04+, Windows 10+, WSL; Linux Ubuntu Server 20.04+,

RESEARCH INTERESTS

- **Computational Mechanics and Applied Mathematics**
- **Shelf-Ice Mechanics and Rheology**
- **Non-linear Large-Deformation Elasticity, Visco-Plasticity, and Elasto-Plasticity**
- **Quasi-Brittle and Ductile Failure Modes**
- **Computational Algorithm Optimization**
- **Data Analysis and Digital Infrastructures**
- **Google Maps APIs and REST APIs**
- **Emerging Financial Technologies**
- **High-Performance and Parallel Computing**

PROFESSIONAL AND ACADEMIC EXPERIENCE

Professional: Chief Innovation Officer and Lead Backend Engineer

ComputeMore Africa Technologies Ltd (Tech Startup) | Blantyre, Malawi

October 2025 – Present | Experience: 5 months

***As Chief Innovation Officer I am tasked with consulting on Systems Design, proposing new projects and the double role of Lead Backend Engineer gives me first-class authority over Systems Architecture, Quality Control, Implementation and DevOps**

- **Achievements:**
 - Developed the Ecoride App API (a new ride-sharing app in Malawi looking to implement interesting Financial Technologies with Mobile Money Airtel and TNM). **Stack:** Flutter, Vite Node.js JavaScript, Redis PostgreSQL, C# .NET ASPCORE, Google Maps API, PayChangu API, RabbitMQ and SignalR WebSocket
 - Developed new app called INTEGRATE for a Hotel ERMS (Enterprise Resource Management System) – **Stack:** Flutter, React Typescript, PostgreSQL, Redis, C# .NET ASPCORE, SignalR Websockets
 - Leveraging **Cloudflare R2 Object Storage** for egress-free, low-cost cloud storage
 - Built RestAPIs including Authentication, and Authorisation using **BCrypt** and **JWT** 256-bit tokens for security hardening. Clean architecture following the Controller -> DTOs -> Services -> Entities philosophy.
 - **DevOps** managed with MS Azure and Render for Databases, Vercel for the web admin portals and our company website.
 - Our tech partners included Namecheap, Cloudflare for web hosting and blob/cloud storage (R2); PayChangu (Fintech provider), and ICTAM.
 - **(References: [Mr. Thanthwe Kalonga](#))**

Professional: SEAT* Backend Application Developer (Hybrid) – Experience: 1.3 Years

University of Cape Town | Cape Town, South Africa

December 2025 – February 2026 | November 2024 – July 2025 | December 2021 – January 2022

***SEAT is an in-house enrolment platform developed at the Faculty of Engineering & the Built Environment**

Front-end: Python Django with HTML Views. **Database Architecture:** PostgreSQL (psycopg2) - **Remote Server-Hosted.**

- **Tasks and Achievements:**
 - **DevOps** managed with Git CLI, Linux BASH Shell (Ubuntu 24.04 Server) and Django CLI/Shell tools.
 - Code inspection, debugging, re-factoring and module linking (Python Django, JavaScript, HTML)

- Data Engineering – managing database migrations, database updates, database access control and database resource management (PostgreSQL, pgAdmin, Django ORM)
- Developed an AI-based timetable generator
- Improved server availability to 100% with 0 server errors in a month
- Maintained client (end-user) verification and authorization protocols with LDAP
- Some UX design also was done, implementing new components to make it easier for students to see their outstanding enrolments
- Email automation with SMTP and Django's shell-based direct code-execution
- (References: [Prof. Sebastian Skatulla](#), [Ms. Celeste Wilson](#))

Academic: **SESKA* Computational Engineer** as part of **Doctor of Philosophy: Civil Engineering**

University of Cape Town | Cape Town, South Africa

September 2023 – Present | Experience: 2.5 Years

***SESKA is an in-house Finite Element Analysis software package maintained in the Department of Civil Engineering**

Back-End: C++, and Fortran. **HPC and Parallel Computing:** PETSc, OpenMPI, UCT HPC. OS: Linux, Windows

• Tasks and Achievements:

- **DevOps** managed with Git CLI, SmartGit GUI and BitBucket Web.
- Deployed SESKA onto UCT HPC, the EBE Computational Mechanics Labs, and Chemical Engineering Computer Labs with a Full-Cycle development of windows Port of SESKA with UCRT and MVSC++, that I led and completed solely.
- Implemented of working **elasto-plasticity**, **visco-plasticity**, **material damage**, **cyclic loading**, and **variable elasticity** models.
- Innovations include computational improvements using inline ternaries instead of nested conditionals, pre-incrementation for loops, and memory clean-ups for temporary quantities – preferring in-place computing.
- (References: [Prof. Sebastian Skatulla](#), [Mr. Sibonelo Twala](#), [Mr. Theo Armstrong](#), [Dr. Maxime Nutte](#))

Academic: **Assistant Lecturer in Engineering Mechanics** – Experience: 2 Years

University of Cape Town | Cape Town, South Africa

February 2025 – July 2025 | June 2024 – December 2024 | June 2023 – December 2023 | June 2022 – December 2022

• Tasks and Achievements:

- 85%+ Course Pass Rate for CIV2011F (2025), 93%+ Course Pass Rate for CIV1007S (2022, 2023, 2024)
- Crafted the course outlines, course content and exams from Scratch.
- Led and directed a team of five Teaching Assistants, managed budgets and schedules
- Enhanced technical and practical instruction of students using ***SESKA**
- (References: [Prof. Sebastian Skatulla](#), [Ms. Ayesha Dalwai](#))

EDUCATION HISTORY

Ongoing **Doctor of Philosophy: Civil Engineering**

University of Cape Town | Cape Town, South Africa | September 2023 – Present | Expected Completion August 2026

- **Topic of study** – *A Nonaffine Damage Theory for Multisolid Composites*
- **Research Keywords:** Composite Materials, Viscoelasticity, Computational Mechanics, Applied Mechanics
- **Skills Developed:** Research Management, Research Design and Reporting, HPC, Microsoft Visual Studio C++, GNU C++, GNU Fortran, Mingw32/64 Cross-Compatibility Tools, Python for Data Science, Git, MSYS2 Shells and Linux BASH.

Master of Science in Engineering: **Structural Engineering & Materials**

University of Cape Town | Cape Town, South Africa | March 2020 – December 2021

- **Distinction in Thesis** – *Towards a material damage model using the logarithmic strain, with von Mises plasticity considerations.*
- **Research Keywords:** Material Damage, Computational Mechanics, Applied Mechanics
- **Skills Developed:** Research Management, Research Design and Reporting, HPC, GNU C+, Python for Data Analysis, Git and Linux BASH.

Bachelor of Science (Honours): **Civil Engineering**

University of Cape Town | Cape Town, South Africa | January 2016 – December 2019

- **Distinction in Dissertation** – *Design of A Climate Chamber for Uni-axial Compressive Strength Testing of Laboratory Grown Sea Ice.*
- **Dean's Merit List:** 2016, 2017.

LICENSES & CERTIFICATES

Ongoing **Professional Certificate: IBM Data Engineering**

Coursera | July 2025 – Present | Expected Completion End February 2026 | 8/16 Modules Completed (150+ hours)

- IBM – Introduction to Data Engineering
- IBM – Python for Data Science, AI & Development

- IBM – Python Project for Data Engineering
- IBM – Introduction to Relational Databases (RDBMS)
- IBM – Databases and SQL for Data Science with Python
- IBM – Hands-on Introduction to Linux Commands and Shell Scripting
- IBM – Relational Database Administration (DBA)
- IBM – Introduction to Big Data with Spark and Hadoop

Ongoing Professional Certificate: Microsoft Back-End Developer

Coursera | September 2025 – Present | Expected Completion End April 2026 | 3/8 Modules Completed (40+ hours)

- Microsoft – Foundations of Coding Back-End
- Microsoft – Security and Authentication
- Microsoft – Database Integration and Management

Professional Certificate: Google Advanced Data Analytics

Coursera | December 2024 – May 2025 | 300+ hours of training

- Google - Foundations of Data Science
- Google - Regression Analysis: Simplify Complex Data Relationships
- Google - The Power of Statistics
- Google - Go Beyond the Numbers: Translate Data into Insights
- Google - Get Started with Python
- Google - The Nuts and Bolts of Machine Learning

Developer Certificate: Data Analysis with Python

Coursera | April 2025 – May 2025 | 300+ hours of training

- Data Structures: Pandas dataframes, Numpy Arrays
- Data Cleaning and Visualization: Matplotlib, Seaborn
- Data Scraping and Data Extraction: .csv, HTML, BigQuery

ORGANIZATIONS AND TECH SUBSCRIPTIONS

- GitHub Pro Contributor (Open-Source and Closed-Surface) – 500+ contributions in last 6 months – <https://github.com/paulnamalomba>
- PyPI Contributor (kabwenzenamalomba@gmail.com – diversecellar)
- AfricasTalking API Subscription (kabwenzenamalomba@gmail.com), Airtel Africa API Developer Platform (kabwenzenamalomba@gmail.com)
- Upstash Context7 API Subscription (kabwenzenamalomba@gmail.com)
- Android Studio Developer Platform (kabwenzenamalomba@gmail.com); Google Cloud Developer Platform (kabwenzenamalomba@gmail.com); Google AI Pro Subscription (kabwenzenamalomba@gmail.com)
- South African Statistical Association, July 2025 – Present [ID 2748543]
- Statistics Without Borders, January 2025 – Present [ID 16089]
- South African Institution of Civil Engineering, 2021 – Present [ID 202100183]

CONFERENCES & PUBLICATIONS

- MSc Thesis 2022 [available: <http://hdl.handle.net/11427/37632>]: *Towards a material damage model using the logarithmic strain, with von Mises plasticity considerations.*
- Africomp7 conference, February 2026 – scheduled to present two abstracts:
 - A Lie Group Approach to Non-Affine Interaction and Damage in Multi-Solid Composites with Prof. Sebastian Skatulla, Prof. Michael Kaliske, Prof. Carlo Sansour
 - Logarithmic Strain-Based Glen's Flow Law for Glaciological Applications with Prof. Sebastian Skatulla, Prof. Michael Kaliske, Prof. Carlo Sansour, Dr. Maxime Nutte
- ICMM7 Conference, November 2022.
- SACAM2020 Conference, November 2021.

COMMUNITY INVOLVEMENT & LEADERSHIP

- CRC Cape Town (Malawi): Service in Altar Team, July 2025 – Present.
- Redeemed For a Purpose Ministries (Malawi): Service in Financial Coordination, April 2025 – Present.
- International Students Forum (UCT): March 2025 – Present.
- Statistics Without Borders Volunteer, January 2025 – Present.
- Mentorship and Supervision of Honour's Researchers (UCT), June 2021 – Present.
- Volunteer Work for IAPO Soccer Tournament (UCT), October 2022.

- **Flood and Disaster Relief Efforts (Mangochi, Malawi), January 2022.**
- **Child Education Advocacy (Mangochi, Malawi), January 2017 & January 2018.**

REFEREES

Prof. Sebastian Skatulla^[1,2] – PhD Research Supervisor, MSc Research Supervisor, SEAT Lead Developer

[1] Professor of Structural Engineering & Materials | Department of Civil Engineering, Faculty of Engineering & the Built Environment, University of Cape Town

[2] Director | Computational Continuum Mechanics Research Group, University of Cape Town

📍 Room 3.41, New Engineering Building, Upper Campus, Cape Town

☎ +27216502595 | ✉ sebastian.skatulla@uct.ac.za

Mr. Theo Armstrong^[1,2] – Laboratory Administrator (UCT ICTS)

Laboratory Administrator | Student Computing Services, University of Cape Town

☎ +27216503464 | ✉ theo.armstrong@uct.ac.za

Mr. Sibonelo Twala^[1,2] – Senior Systems Engineer (UCT ICTS)

Senior Systems Engineer | Workplace Services, University of Cape Town

✉ sibonelo.twala@uct.ac.za

Ms. Celeste Wilson – Deputy Faculty Manager (UCT EBE)

Deputy Faculty Manager | Faculty of Engineering & the Built Environment, University of Cape Town

📍 Level 6 Admin, New Engineering Building, Upper Campus, Cape Town

☎ +27216504194 | ✉ celeste.wilson@uct.ac.za

Ms. Ayesha Dalwai – Departmental Manager (UCT EBE – Civil Engineering)

Departmental Manager | Department of Civil Engineering, Faculty of Engineering & the Built Environment, University of Cape Town

📍 Level 2 Admin, New Engineering Building, Upper Campus, Cape Town

☎ +27216502591 | ✉ ayesha.dalwai@uct.ac.za

Dr. Keith MacHutcheon – MSc Research Supervisor

Honorary Research Associate | Department of Civil Engineering, Faculty of Engineering & the Built Environment, University of Cape Town

📍 Room 3.40, New Engineering Building, Upper Campus, Cape Town

☎ +27216505398 | ✉ keith.machutcheon@uct.ac.za

Dr. Maxime Nutte – Research Collaborator

Postdoctoral Fellow | Department of Civil Engineering, Faculty of Engineering & the Built Environment, University of Cape Town

📍 Level 3 Postdoc Research Space, New Engineering Building, Upper Campus, Cape Town

☎ +33633800414 | ✉ maxime.nutte@uct.ac.za

Mr. Thanthwe Kalonga^[1,2] – Managing Director of ComputeMore Africa Technologies Ltd

[1] Executive Managing Assistant and Actuarial Manager | NICO Life (Sanlam-Allianz)

[2] Managing Director | ComputeMore Africa Technologies Ltd

📍 PO Box 31548, Chichiri 3, Blantyre, Malawi

☎ +265987629744 | ✉ thanthwekalonga7@gmail.com | ✉ thanthwe.kalonga-md@computemore.com